

Some Discussion for a tyve of Generalized Heisenberg Lie Algebra

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Abstract

In this note, we study a class of step two nilpotent Lie algebras arising from a linear space equipped with a constant Poisson structure, which could be realized as a generalized Heisenberg Lie algebra. We discuss the structure of these algebras, associating the singular point of its coadjoint orbit to the affine variety.

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1 Preliminary

In this section, we briefly review some basic notion of Poisson manifolds, and discuss some properties of constant Poisson structure on a finite dimensional linear space.

1.1 Poisson Manifold and Hamiltonian vector field

Definition 1.1 (Poisson manifold). A Poisson manifold $(\mathcal{M}, \{\cdot, \cdot\})$ is a smooth manifold $\mathcal{M}$ equipped with a bilinear operation $\{\cdot, \cdot\} : \mathcal{C}^\infty(\mathcal{M}) \times \mathcal{C}^\infty(\mathcal{M}) \rightarrow \mathcal{C}^\infty(\mathcal{M})$, called the Poisson bracket, satisfying the following properties:

1. Skew-symmetry: $\{f, g\} = -\{g, f\}$ for all $f, g \in \mathcal{C}^\infty(\mathcal{M})$.
2. Jacobi identity: $\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0$ for all $f, g, h \in \mathcal{C}^\infty(\mathcal{M})$.
3. Leibniz rule: $\{f, gh\} = g\{f, h\} + h\{f, g\}$ for all $f, g, h \in \mathcal{C}^\infty(\mathcal{M})$.

Definition 1.2 (Hamiltonian vector field). Given a Poisson manifold $(\mathcal{M}, \{\cdot, \cdot\})$ and a smooth function $H \in \mathcal{C}^\infty(\mathcal{M})$, the Hamiltonian vector field X_H associated with H is defined by the relation

$$\langle df, X_H \rangle = \{f, H\}, \quad \forall f \in \mathcal{C}^\infty(\mathcal{M}),$$

where $\langle \cdot, \cdot \rangle$ denotes the natural pairing between differential forms and vector fields.

1.2 Constant Poisson structure on a linear space

Definition 1.3 (Constant Poisson structure). Let V be a finite dimensional linear space over $\mathbb{R}$, and let $\{\cdot, \cdot\}$ be a Poisson bracket on V . We say that the Poisson structure is constant if the Poisson bracket of any two linear functions on V is constant. In other words, for any linear functions $\xi, \eta \in V^*$, we have $\{\xi, \eta\} \in \mathbb{R}$.

Since the bracket with any constant function is zero, the bracket actually defines a skew-symmetric bilinear form Ω on the dual space V^* by $\Omega(\xi, \eta) := \{\eta, \xi\}$. One can check that the Jacobi identity is automatically satisfied for a constant Poisson structure, and thus the Poisson bracket is completely determined by the skew-symmetric bilinear form Ω on V^* once we identify the tangent space $T_v V$ of V at any point $v \in V$ with V itself and its double dual V^{**} , and also the differential $d_v \xi$ of any linear function $\xi \in V^*$ with ξ itself. Note that the Hamiltonian vector field X_ξ is given by $\Omega_{\sharp}(\xi) := \iota_{X_\xi} \Omega \in V$, which is a constant vector field on V .

Next we restrict our attention to the space of all symmetric bilinear forms on V , denoted by $S^2(V)$. For $\sigma \in S^2(V)$, write $\sigma(v) = \frac{1}{2}\sigma(v, v)$ as a smooth function on V . Note that $d_v \sigma = \iota_v \sigma \in T_v^* V \cong V^*$.

We first compute the bracket of two quadratic σ_1, σ_2 :

$$\begin{aligned} \{\sigma_1, \sigma_2\}(v) &= \langle d_v \sigma_1, X_{\sigma_2}(v) \rangle = \langle \iota_v \sigma_1, X_{\sigma_2}(v) \rangle = \{\iota_v \sigma_1, \sigma_2\}(v) \\ &= -\{\sigma_2, \iota_v \sigma_1\}(v) = -\langle d_v \sigma_2, X_{\iota_v \sigma_1}(v) \rangle = -\langle \iota_v \sigma_2, X_{\iota_v \sigma_1}(v) \rangle \\ &= -\{\iota_v \sigma_2, \iota_v \sigma_1\} = \{\iota_v \sigma_1, \iota_v \sigma_2\} \end{aligned}$$

We first compute the bracket of a quadratic function σ and a linear function ξ :

$$\{\sigma, \xi\}(v) = \langle d_v \sigma, X_\xi \rangle = \sigma(v, X_\xi) = \langle \iota_{X_\xi} \sigma, v \rangle. \quad (1)$$

The formula above shows that the bracket $\{\sigma, \xi\}$ is a linear function.